

Reviewer Pack — Minimal Rank of Quasi-Frobenius Algebras and DPLL Proof Width...

Entry b3173a582d9e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quasi-Frobenius Algebras and DPLL Proof Width Correlation

Entry ID: b3173a582d9e **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-03 05:38:26 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Algebra (Quasi-Frobenius Algebras) **Field B** (complexity object): DPLL Search Trees

Statement:

For every Boolean satisfiability problem φ , the minimal rank of a quasi-Frobenius algebra associated with φ is linearly correlated with the DPLL proof width $w(\varphi)$, such that the minimal rank of the quasi-Frobenius algebra is $\Theta(w(\varphi))$.

Rationale (proposer's reasoning):

Quasi-Frobenius algebras have been used in noncommutative geometry and algebraic combinatorics, offering a rich algebraic structure that could potentially encode the complexity of computational problems. The DPLL proof width is a standard measure of the difficulty of a problem in the context of DPLL-based SAT solvers. A link between these two areas may expose new insights into the nature of computational hardness.

Taxonomy category: Noncommutative Algebra (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e98a3ab9b910cfb2

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for at least 80% of the generated Boolean formulas φ , the correlation coefficient between the minimal rank of the quasi-Frobenius algebra and DPLL proof width $w(\varphi)$ exceeds 0.7, with $p\text{-value} \leq 0.05$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - intitle:Quasi-Frobenius Algebras AND DPLL proof width - title:(Minimal Rank OR Quasi-Frobenius Algebras) AND DPLL search trees - subject:DPLL AND quasi-Frobenius algebra AND proof width

Top relevant hits considered: - [http://arxiv.org/abs/1708.04361v3] A short proof that the free associative algebra is Hopfian - [http://arxiv.org/abs/2401.03308v3] On von Neumann regularity of ample groupoid algebras - [http://arxiv.org/abs/1802.07219v3] Leibniz algebras as non-associative algebras

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def generate_cnf(n, m):
    cnf = []
    for _ in range(m):
        clause = [random.randint(-n, -1), random.randint(1, n)]
        cnf.append(clause)
    return cnf

def dpll(cnf, assignment={}):
    if not cnf:
        return True
    unit_clause = next((c for c in cnf if len(c) == 1), None)
    if unit_clause:
        lit = unit_clause[0]
        if lit in assignment and assignment[lit] != (lit > 0):
            return False
        return dpll([c for c in cnf if lit not in c], {**assignment, lit: True})
    pure_literal = next((l for l in range(1, n + 1) if all(l not in c or -l not in c for c in cnf)), None)
    if pure_literal:
        return dpll([c for c in cnf if pure_literal not in c], {**assignment, pure_literal: True})
    lit = random.choice(cnf[0])
    return dpll(cnf + [[-lit]], {**assignment, lit: True}) or dpll(cnf + [[lit]], {**assignment, lit: False})

def quasi_frobenius_rank(cnf):
    n = len(set(abs(lit) for lit in cnf))
    if n == 0:
        return 0
```

```

rank = 0
while True:
    assignment = {}
    for lit in range(1, n + 1):
        if random.choice([True, False]):
            assignment[lit] = True
        else:
            assignment[-lit] = True
    if dpll(cnf, assignment):
        rank += 1
    else:
        return rank

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    min_ranks = []
    proof_widths = []

    for n in n_values:
        for _ in range(5): # Ensure at least 30 instances per seed
            cnf = generate_cnf(n, random.randint(1, n * 2))
            rank = quasi_frobenius_rank(cnf)
            min_ranks.append(rank)
            proof_widths.append(len(dpll(cnf)))

    if not min_ranks or not proof_widths:
        return {
            "metric_name": "Correlation Coefficient",
            "metric_value": 0.0,
            "instances_tested": len(min_ranks),
            "n_max": max(n_values),
            "conjecture_holds": False,
            "counterexample": "empty_data"
        }

    mean_rank = sum(min_ranks) / len(min_ranks)
    mean_width = sum(proof_widths) / len(proof_widths)
    covariance = sum((min_ranks[i] - mean_rank) * (proof_widths[i] - mean_width) for i in range(len(min_ranks)))
    variance_rank = sum((min_ranks[i] - mean_rank) ** 2 for i in range(len(min_ranks))) / len(min_ranks)
    variance_width = sum((proof_widths[i] - mean_width) ** 2 for i in range(len(proof_widths))) / len(proof_widths)
    correlation_coefficient = covariance / (math.sqrt(variance_rank) * math.sqrt(variance_width))

    return {
        "metric_name": "Correlation Coefficient",
        "metric_value": correlation_coefficient,
        "instances_tested": len(min_ranks),
        "n_max": max(n_values),
        "conjecture_holds": abs(correlation_coefficient) >= 0.7,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97]
    results = []

```

```

for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results) / len(results)
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction=1.0")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample='correlation_coefficient_too_low' first_failing_s

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_25ac1e16.py", line 102, in <module>
    result = run_trial(seed)
              ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_25ac1e16.py", line 66, in run_trial
    rank = quasi_frobenius_rank(cnf)
           ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_25ac1e16.py", line 41, in quasi_frobeni
    n = len(set(abs(lit) for lit in cnf))
           ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_25ac1e16.py", line 41, in <genexpr>
    n = len(set(abs(lit) for lit in cnf))
               ~~~~~
TypeError: bad operand type for abs(): 'list'

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete the required number of trials to meet the pre-registered support condition for the conjecture. | next: Investigate and fix the error in the test code to ensure it can run to completion and provide the necessary data for further analysis.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17637
2	preregistration	ollama_remote	glm4:latest	0	0	13271
3	novelty	ollama_remote	glm4:latest	0	0	20977
4	novelty	ollama_remote	glm4:latest	0	0	12280
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12718
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22074
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19143
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16429
9	judge	ollama_remote	glm4:latest	0	0	22525

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 157055 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/b3173a582d9e.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/b3173a582d9e.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/b3173a582d9e.tar.gz` (if generated)